

Submission LINALG1-MID-SUB02

Question LINALG1-MID-Q01

work / setup:

1. $\det = 2(3) - 4(2) = -2$

2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe): $x=1$, $y=2$ plus 1

Question LINALG1-MID-Q02

work / setup:

1. $\det(A) = 6 \cdot 4 - 2 \cdot 2 = 20$

crossed-out arithmetic: [smudged]

FINAL (maybe): $\det=20$; invertible plus 1

Question LINALG1-MID-Q03

work / setup:

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear plus 1

Question LINALG1-MID-Q04

work / setup:

1. Set $c \blacksquare (1, 0) + c \blacksquare (0, 1) = (0, 0)$.

crossed-out arithmetic: [smudged]

FINAL (maybe): only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent plus 1

Question LINALG1-MID-Q05

work / setup:

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.

crossed-out arithmetic: [smudged]

FINAL (maybe): the vectors mapped to zero; exactly the solution space of $Ax=0$ plus 1

Question LINALG1-MID-Q06

work / setup:

1. $\det = 5(5) - 2(2) = 21$

2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe): $x=1$, $y=2$ plus 1

Question LINALG1-MID-Q07

work / setup:

1. $\det(A) = 2 \cdot 2 - 2 \cdot 2 = 0$

crossed-out arithmetic: [smudged]

FINAL (maybe): $\det=0$; not invertible plus 1

Question LINALG1-MID-Q08

work / setup:

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear plus 1